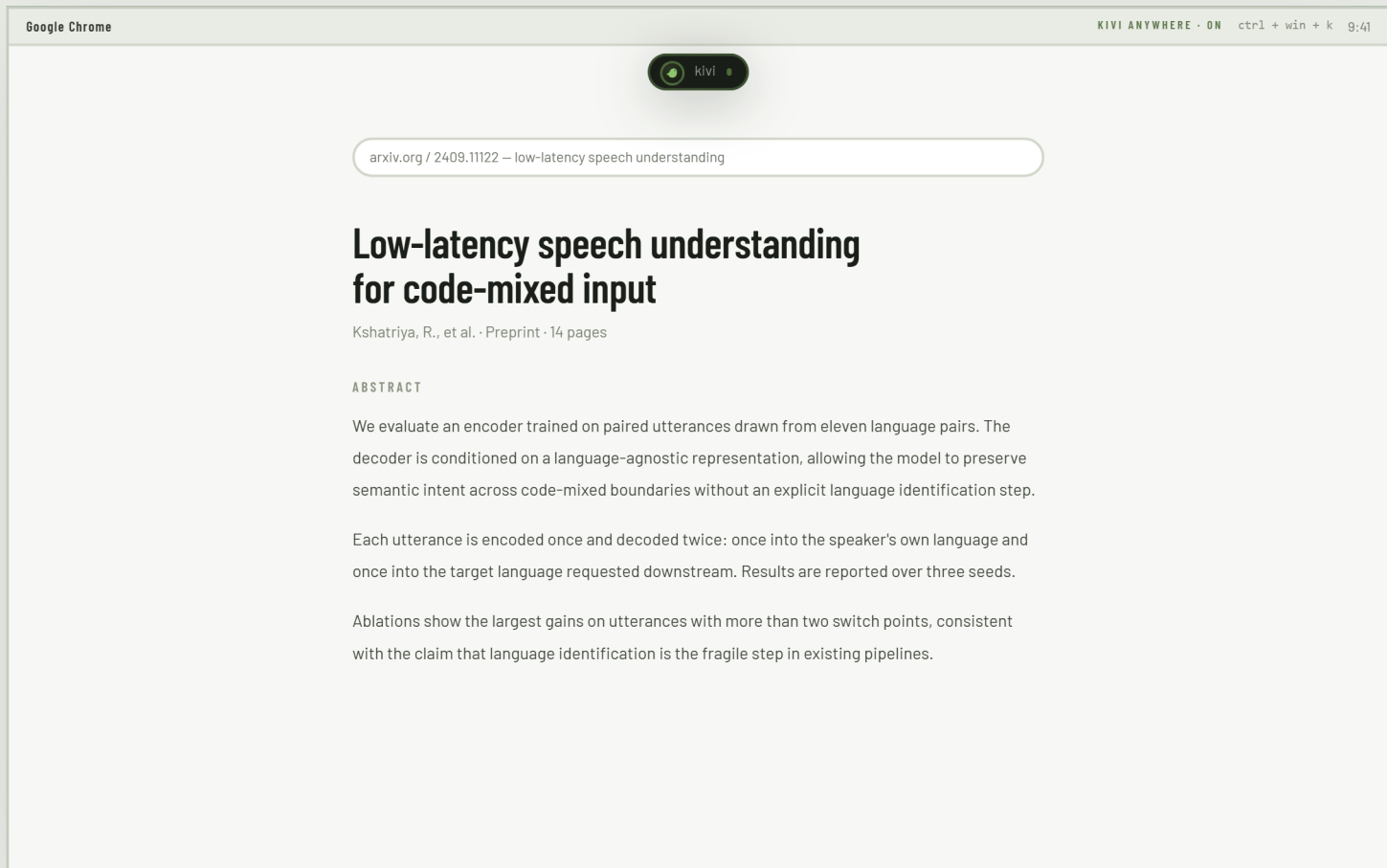


The voice layer, floating over your apps

Twenty-seven screens. In each one the app underneath stays exactly where it was
— Kivi arrives on the selection, takes one instruction, shows a preview, and leaves.

01	02	03	04	05
Point	Speak	Understand	Preview	Apply



02 Green selection

The selection turns green and the panel lands on it, listening. Voice supplies the what.

Google Chrome

LISTENING 9:41

Low-latency speech understanding for code-mixed input

3 · METHOD

Each utterance is encoded once and decoded twice: once into the speaker's own language and once into the target language requested downstream. The implementation requires significant engineering effort and may not be practical within the current timeline.

We therefore report b
on utterances with m

This is consistent with
existing pipelines.

Google Chrome

2 LINES SELECTED

— — — — —

"Explain this like I'm a first-year student."

understand write communicate

listening

ESC TO CLOSE

gest gains

03 Understand

The answer stays beside the paper. Nothing is copied out, nothing is pasted back.

Preview — paper.pdf

9:43

4 · RESULTS

Reported figures are averaged across three seeds with identical schedules.

The decoder is conditioned on a language-agnostic representation, allowing the model to preserve semantic intent across code-mixed boundaries without an explicit language identification step.

We leave joint training with the acoustic front-end to future work.

 Understand

FROM SELECTION

"Explain this like I'm a first-year student."

The model doesn't first decide which language you spoke. It converts your speech into a meaning-level representation that isn't tied to any one language, then writes the output in whichever language is asked for. Skipping the language-guessing step is what removes the delay.

ALSO

simpler still

what's the evidence?

in Tamil



READ ALOUD

KEEP AS NOTE

04 Write · preview

A diff of the selected sentence only. Nothing changes in the document until Apply.

Docs — Method note

9:52

Method note — draft 3

The implementation requires significant engineering effort and may not be practical within the current timeline, so we have decided to report both the full system and a reduced variant instead.

Each run uses the

Remaining open

main objection.

Write

1 SENTENCE

"Make this more concise but keep the technical meaning."

- The implementation requires significant engineering effort and may not be practical within the current timeline, so we have decided to report both the full system and a reduced variant instead.

+ The implementation is costly and may not fit the current timeline; we report both the full system and a reduced variant.

● applies to the selected sentence only

⏮

DISCARD

SHORTER

APPLY

05 Think

Two minutes of messy Tanglish, sorted into decisions, questions and tasks.

Notion — Kivi / working notes

18:07

Private

Working notes

Research log

Prototype v1

People

Working notes — 4 Sep

Insert here

Think

2:14 SPOKEN · TA-EN

"Okay prototype-a finish pannanum, then Arjun-kitta research pathi pesanum, and I'm not sure whether we should use Figma or Framer..."

DECISIONS

Ship the prototype before the research review.

OPEN QUESTIONS

Figma or Framer for the interactive build?

TASKS · PEOPLE

Finish prototype · Talk to Arjun about the research

style · research notes

INSERT INTO PAGE

06 Communicate

One intention, three audiences. Spoken in Tanglish, written in the register you pick.

Gmail – New message

KIVI MIC 21:12

NEW MESSAGE

To

prof.raman@univ.edu

Subject

Absence on Friday

Dear Professor Raman,

I will not be able to attend Friday's session as I have a family commitment that day. I have arranged to go through the material with a classmate and will complete the exercise before the next session.

Thank you for understanding.

Meera

SEND

Communicate

TA-EN → EN

"Na professor-ku mail anuppanum, Friday varamudiyadhu, family function – but formal-ah irukanum, excuse-ah sound aagakoodadhu."

AUDIENCE

professor

teammate

friend

Dear Professor Raman, I will not be able to attend Friday's session as I have a family commitment that day. I have arranged to go through the material with a classmate...

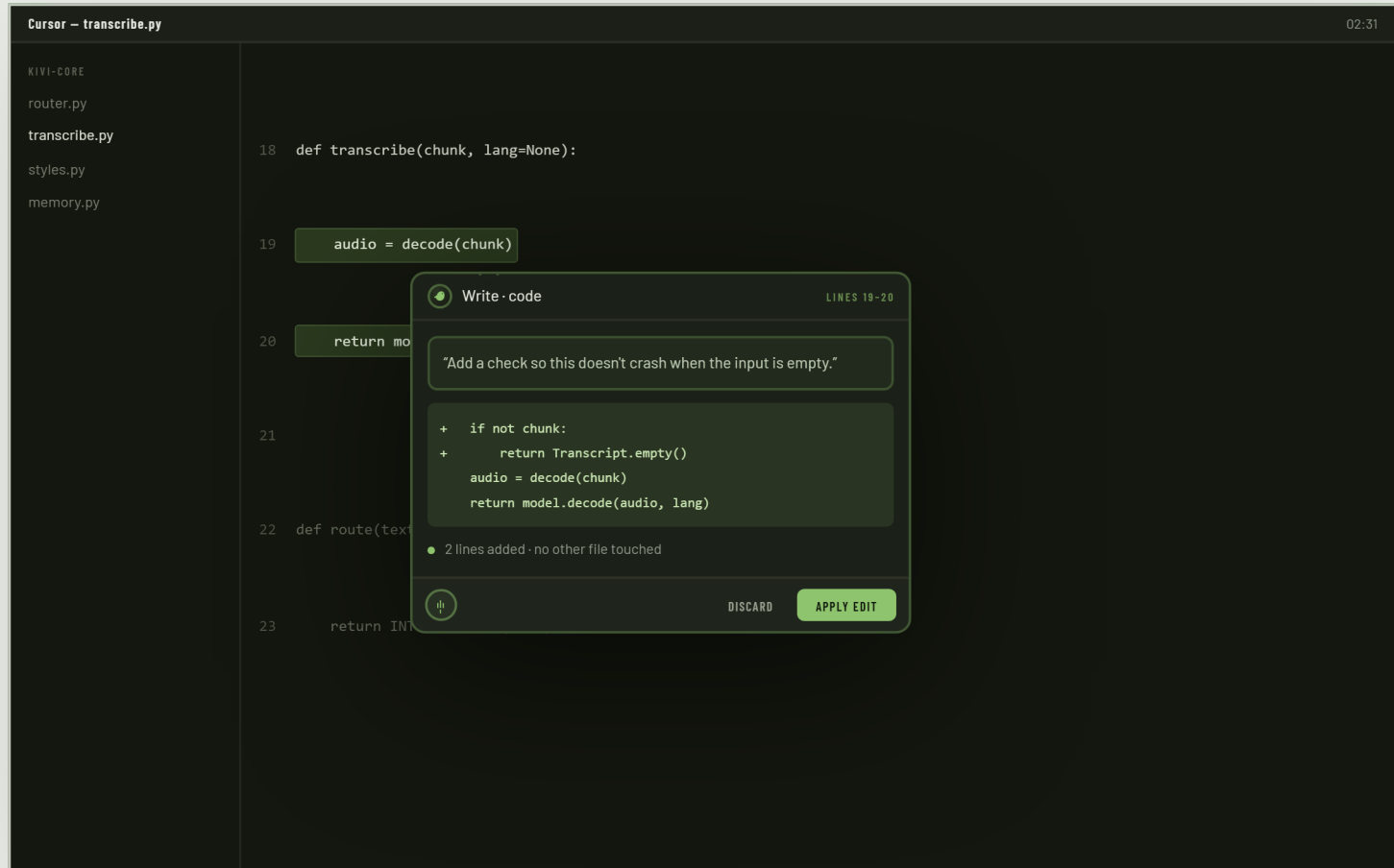
drafted, not sent

WARMER

PLACE IN DRAFT

07 Write · code

A spoken change to the selected lines. Kivi proposes; the developer applies.



08 Remember

Said once in passing, confirmed in place, and returned at nine.

Slack — #design-research

20:48

CHANNELS

design-research

general

eng

Rahul

Arjun · 20:41

I'll send the latency numbers tomorrow morning once the third seed finishes.

Meera · 20:47

Perfect — I'll fold them into the deck before the review.

Message #design-research

Remember

CONFIRM

"Remind me to tell Rahul about the event at 9 PM."

WHEN

Today · 21:00

WHO

Rahul

WHAT

Tell him about the event

Also remembered about you: you prefer concise messages

CHANGE TIME

SET REMINDER

KIVI · 21:00

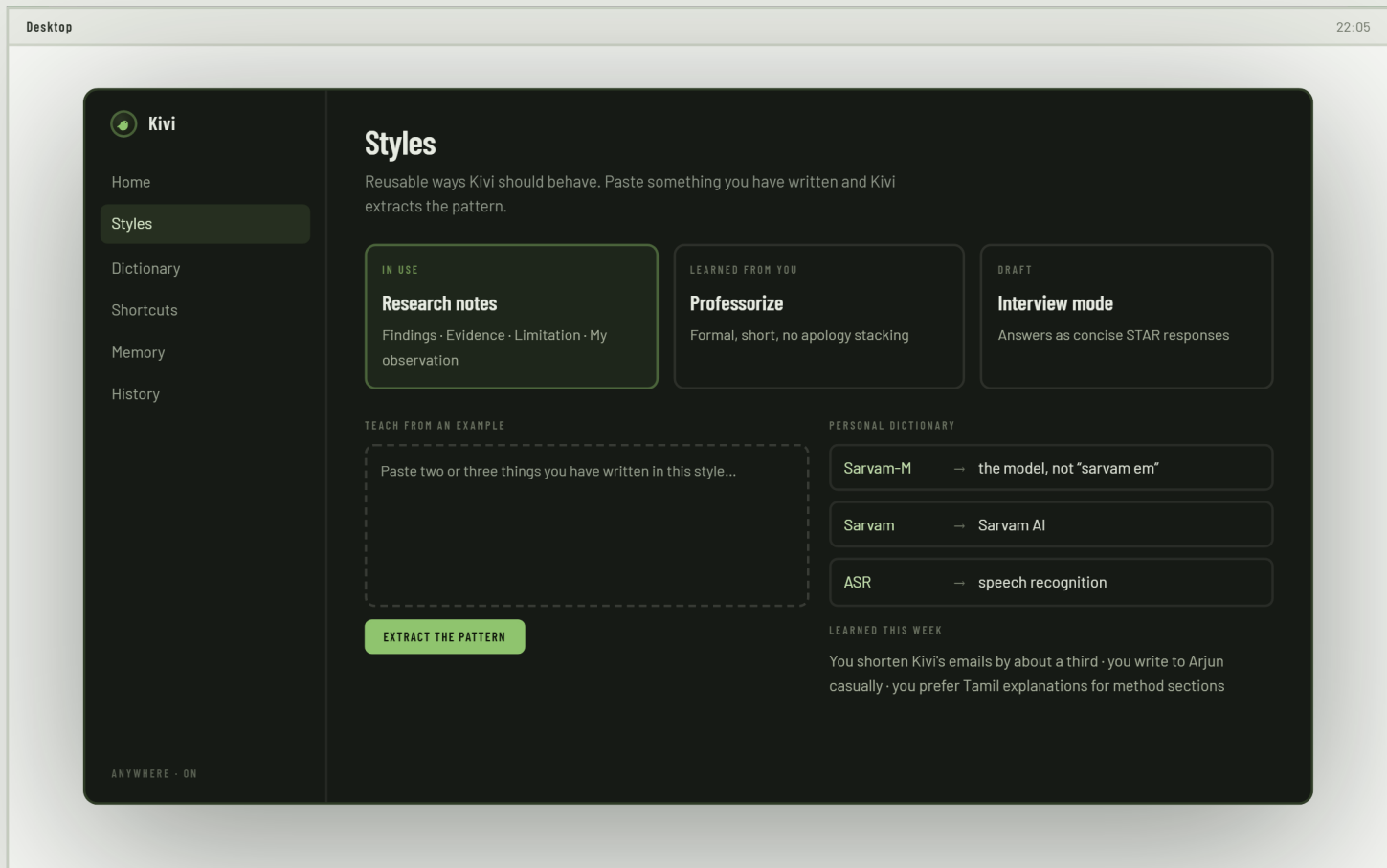
Tell Rahul about the event.

snooze

draft the message

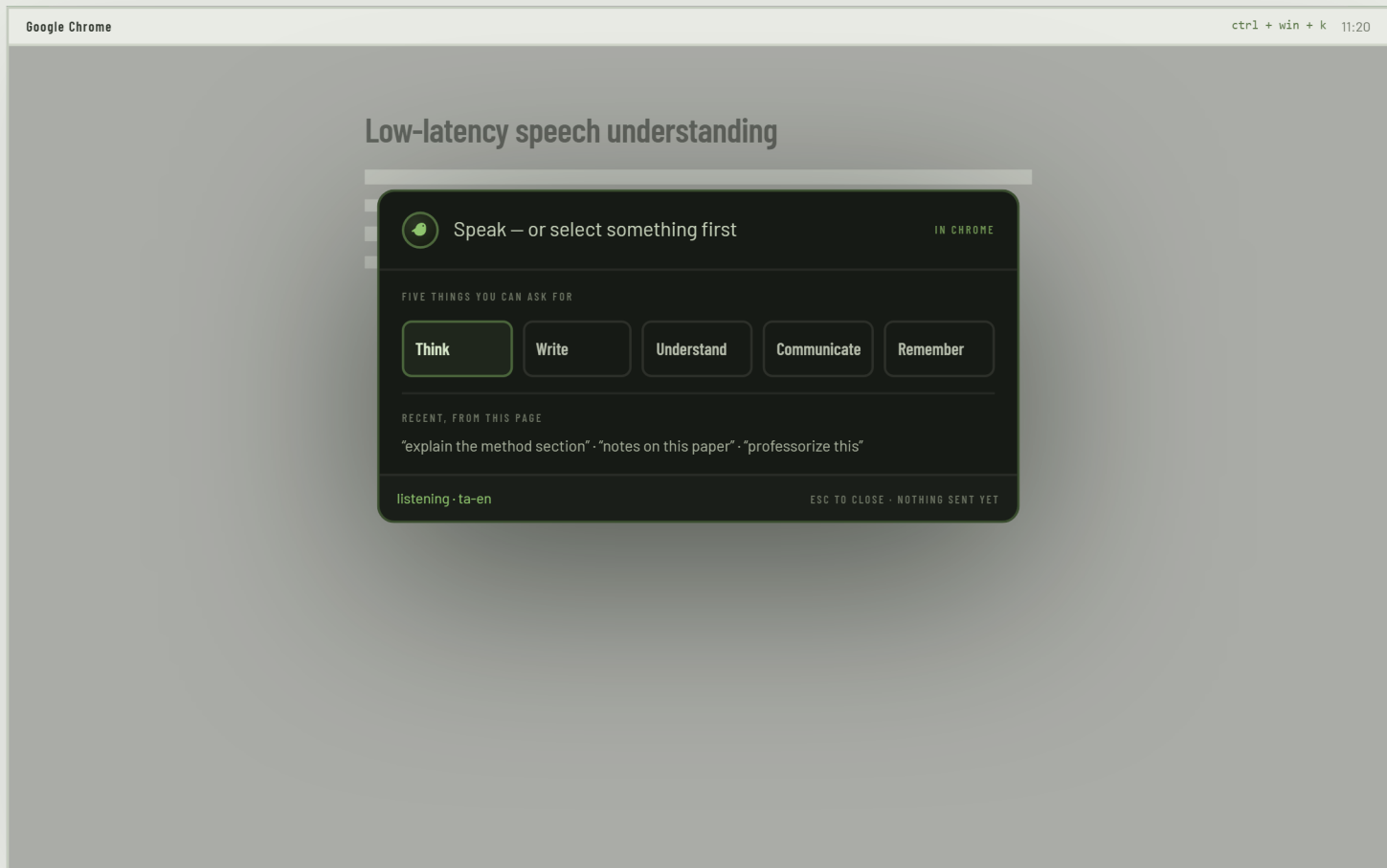
09 Teaching Kivi

The one full window in the product: styles, dictionary and memory in one place.



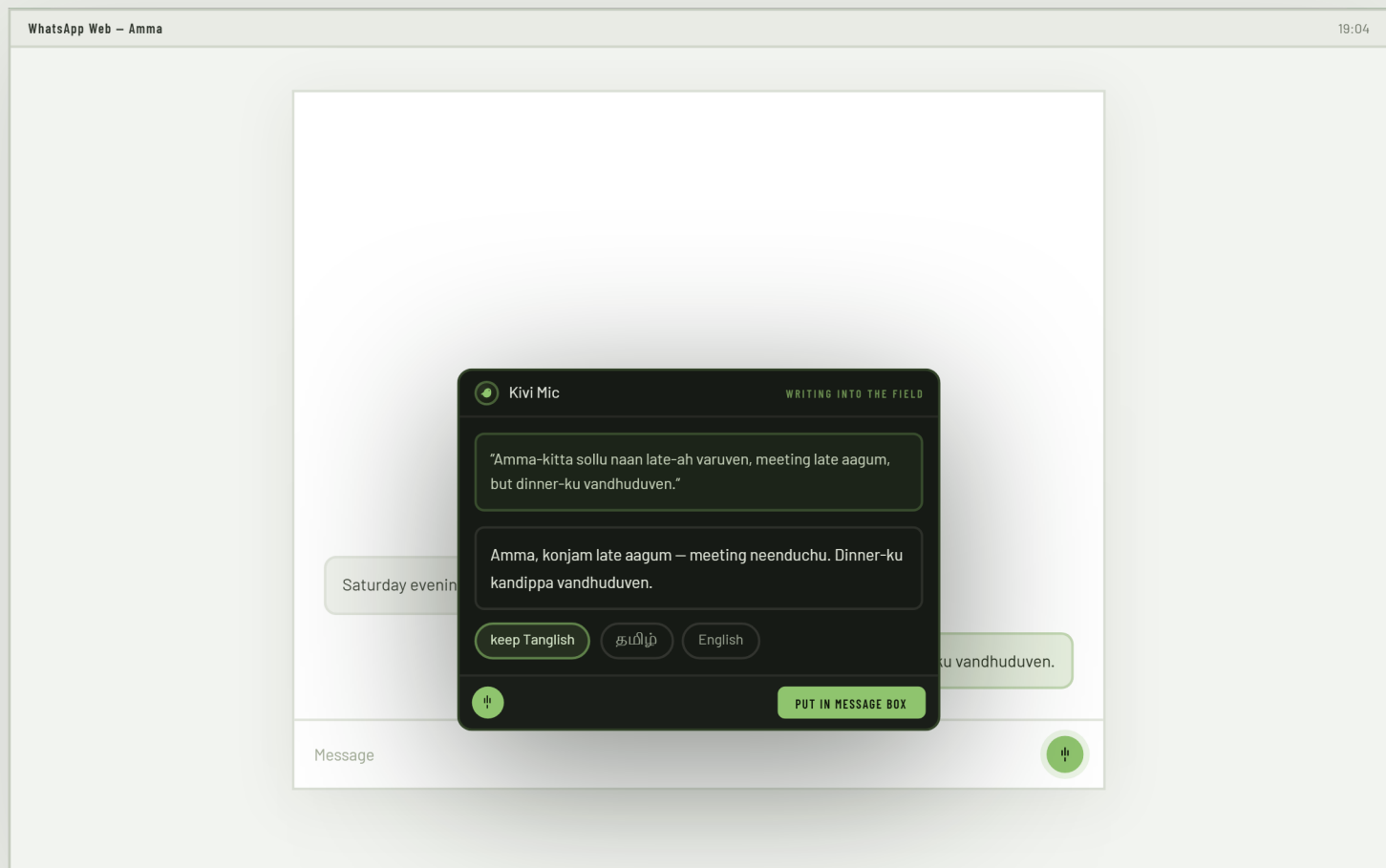
10 Kivi Anywhere

Entry point one. Ctrl + Win + K with nothing selected — Kivi still knows where you are.



11 Kivi Mic

Entry point two. The app's own mic button becomes Kivi, so it writes meaning instead of words.



12 Voice → prompt

Kivi doesn't answer here. It writes the question the other AI can work with.

ChatGPT – new chat13:38

CHATS

New chat

Latency budget

Seed variance

What are you working on?

Ask a question

Write · prompt0:22 SPOKEN

"I need to compare these two approaches but mainly on cost and how hard it is to build, we're only three people, and I want a table and then tell me which one."

kept two axes · table then recommendation

added team size as a constraint

dropped filler, repetition

ADD CONTEXT

PLACE IN BOX

Compare on-device and server-side inference for a code-mixed speech model. Focus on two axes only: cost per hour of audio, and implementation difficulty for a three-person team. Give a short table, then one recommendation with the main risk.

WRITTEN BY KIVI MIC

SEND

13 Understand · code

An unfamiliar function, explained where it sits. Read aloud while you keep scrolling.

Cursor — router.py03:04

```
41 def match(text, table=INTENTS):  
  
42     scored = sorted(  
  
43         ((jaccard(text, k), v) for k, v in table),  
  
44         reverse=True)[:3]  
  
45     return scored[0][1] if scored[0][0] > TAU else None  
  
46  
  
47 def fallback(text):  
  
48     return Intent.CLARIFY
```

Understand · codeLINES 42-45

What is this actually doing?

It scores the spoken text against every intent in the table by word overlap, keeps the best three, and returns the top one – but only if its score clears the threshold `TAU`. Below that it returns nothing, which is what sends the request to `fallback` and asks you to clarify.

why sorted, not max?

explain in Tamil

reading aloud · 0:14

ADD AS COMMENT

14 Voice → structure

Spoken once, arrives as a table with the columns you named.

Docs — Options review

10:16

 Write · structure

APPLIED

"Create a table comparing these four options based on price, difficulty and expected impact."

4 rows · 4 columns

columns taken from your words, in order

Say "add a column for owner" to keep going.



UNDO

ed before the Friday review.

	PRICE	DIFFICULTY	EXPECTED IMPACT
	Low	High	Latency win
	High	Low	Best accuracy
	Medium	Medium	Balanced
Vendor API	Medium	Lowest	Fastest to ship

15 Voice → voice

You speak Tamil, the other person hears Hindi. Neither of you changes app.

Google Meet – Vendor call

15:22

Meera (you)

Vikram · Delhi

Communicate · spoken

தமிழ் → हिन्दी

YOU SAID

உமீவரி டேட்-ஐ ஒரு வாரம் முன்னாடி நகர்த்த முடியுமா?

→

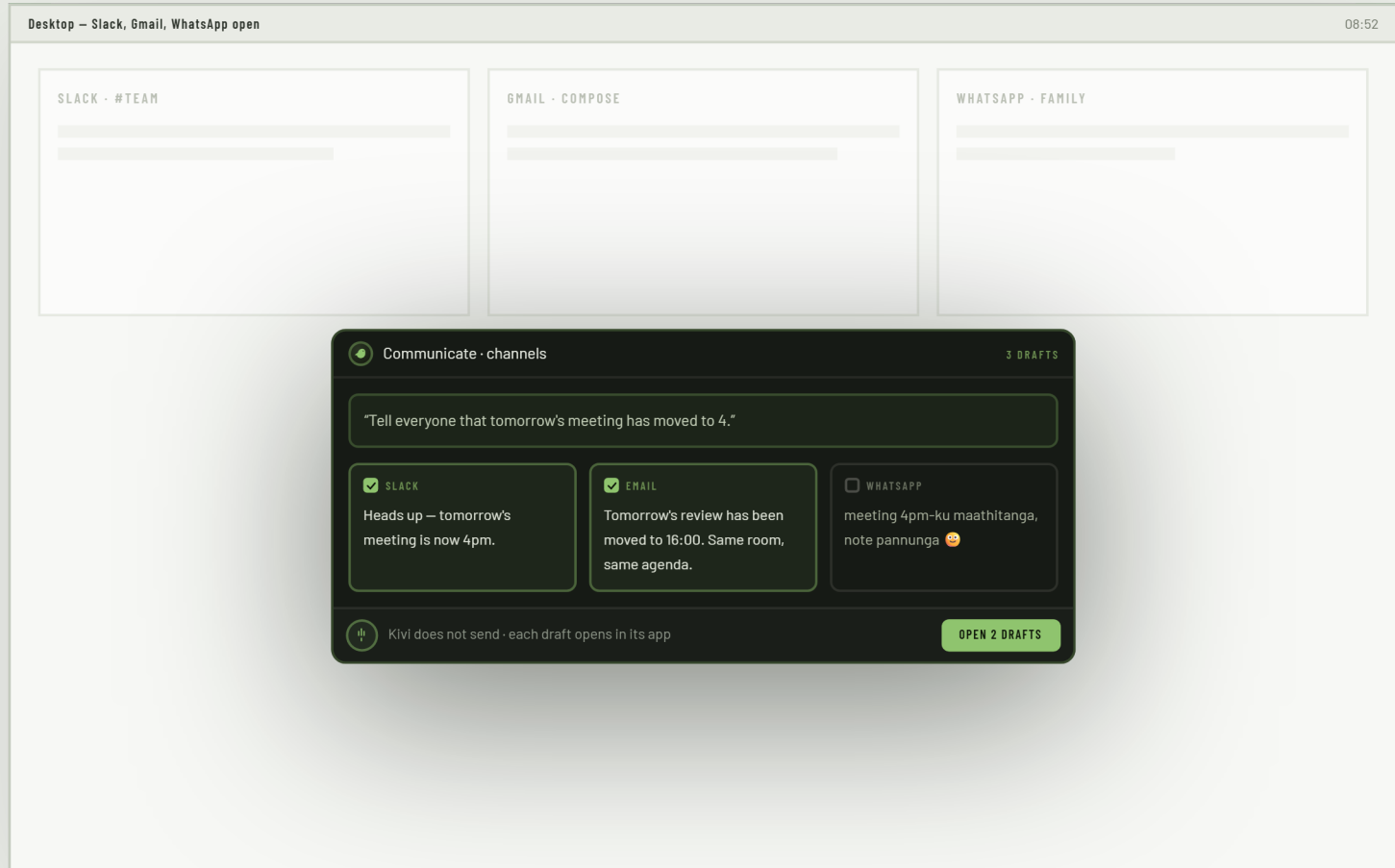
क्या डिलीवरी डेट एक हफ्ते पहले करना possible है?

speaking as you · 1.2s behind

BOTH SIDES SEE THE TRANSCRIPT

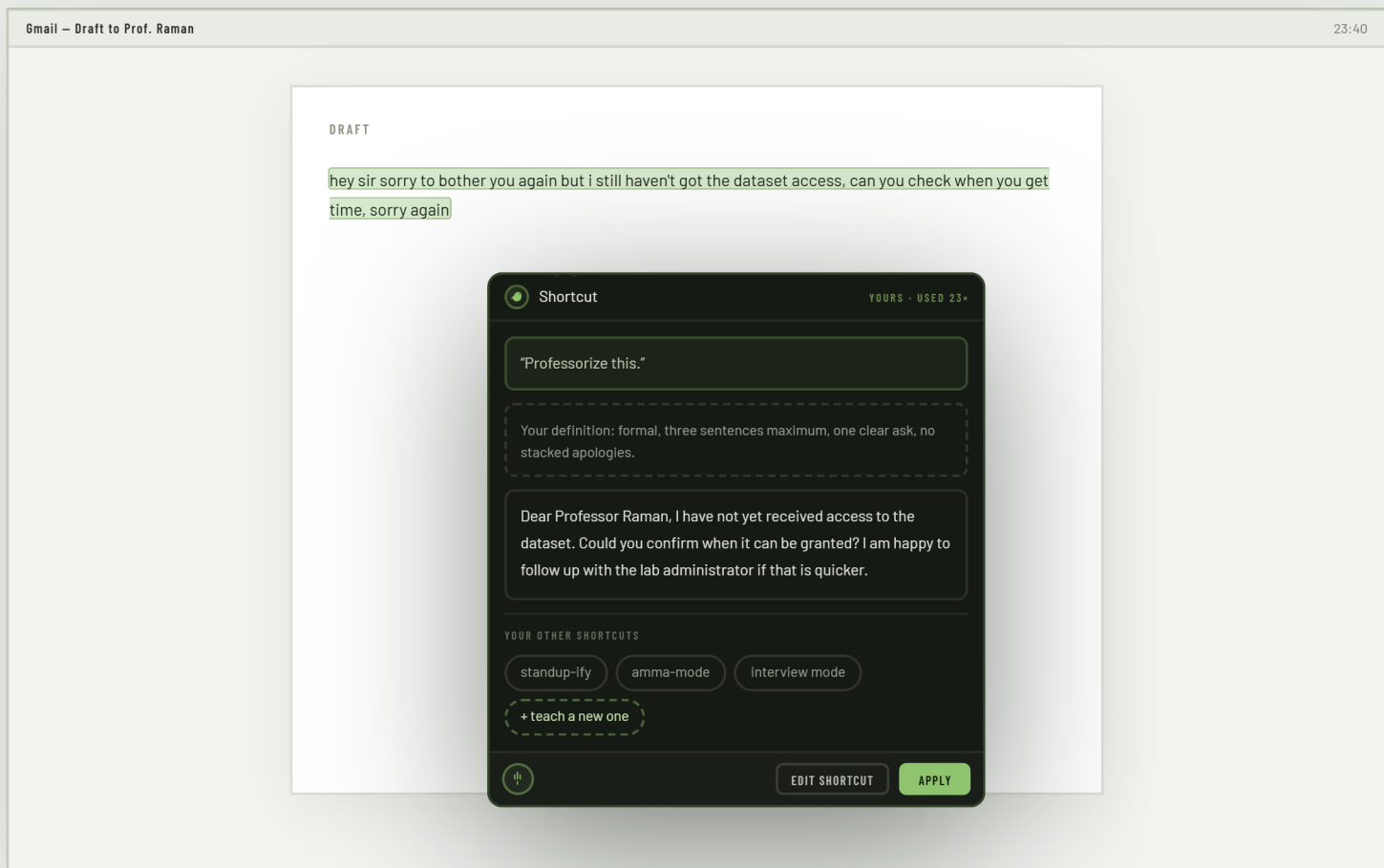
16 One thought, three channels

Kivi writes all three. You choose which ones go, and each one still needs a send.



17 Your own words

A phrase you invented, used as a command — and a whole mode built from your examples.



18 Learning, and limits

Your correction becomes a preference. When Kivi is unsure it asks, and when asked to overstep it declines.

Desktop

mic off 23:58

 Learned from you

You shortened Kivi's last three emails by about a third.

Keep emails to three sentences unless you ask for more?

REMEMBER THIS

NOT A RULE

ALREADY REMEMBERED

Tamil for method sections · casual with Arjun · never "Dear Sir/Madam"

 Not sure yet

LOW CONFIDENCE

"Send this to him."

Two people could be "him" here, and nothing is selected.

Arjun

Rahul

let me select first

What Kivi will not do

- send, post or reply on your behalf
- operate the computer on its own
- listen when the mic is off, or record meetings
- hold a conversation instead of doing the thing

PREVIEW — APPLY — CONFIRM, EVERY TIME

19 Your shortcuts

Every phrase you have given a meaning to, what it does, and where it is allowed to fire.

Kivi

10:12

Kivi

Home

Styles

Dictionary

Shortcuts

Modes

Memory

Shortcuts

7 taught · 141 uses this month

TEACH A SHORTCUT

"professorize this"23×

Formal, three sentences maximum, one clear ask, no stacked apologies.

selectionEmail - Slack

"standup-ify"41×

Yesterday / today / blockers, one line each, no adjectives.

voice onlyanywhere

"amma-mode"18×

Same message in Tamil, warm, no technical words.

selectionWhatsApp

"guard this"31×

Add null and empty-input checks, keep the existing signature.

selectionVS Code

NEEDS YOUR ATTENTION

"short version" sounds close to "shorter version". Kivi guessed wrong twice this week.

RENAME

A shortcut is a phrase you say. A mode is how Kivi behaves until you turn it off.

20 Teach a shortcut

Say the phrase, say what it should mean. Kivi writes the rule back in your words for you to correct.

Kivi — Shortcuts

LISTENING 10:14

New shortcut

STEP 2 OF 3

01 · DONE

02 · NOW

03

The phrase

What it means

Try it once

PHRASE

"sarvam-ify"

heard 3× to confirm pronunciation

YOU SAID

"Idha oru product update maadhiri maathu — English la, three bullets, metrics first, no fluff."

KIVI'S READING OF IT

edit any line

01 Rewrite the selection as a product update in English.

02 Exactly three bullets, numbers in the first bullet.

03 No adjectives, no closing line.

FIRES ON

a selection

plain voice

everywhere

Slack · Notion

say more to refine it

BACK

TRY IT ONCE

21 Modes

A shortcut is one command. A mode is how Kivi behaves until you turn it off — and it says so on every panel.

Kivi — Modes

INTERVIEW MODE · ON 17:05

Modes

One at a time. Off is the normal Kivi.

● Interview mode

on since 16:40 · 9 answers

Turns a rambling spoken answer into a STAR response. Keeps your own examples and numbers, cuts hedging, ends on the result.

TURN OFF

Research mode

Everything becomes Finding / Evidence / Limitation / My observation. Never merges your view with the paper's.

yours · 62 uses

Deep work

No suggestions, no follow-up chips. Kivi answers once and closes.

yours · 14 uses

Tamil first

Explanations come back in Tamil, written output stays English.

yours · 27 uses

HOW A MODE TURNS ON

Say it.

"Interview mode on."

Pin it to an app.

Research mode whenever a PDF is in front.

Pick it on the panel.

The mode name sits under the mic while it is on.

22 Build your own mode

Describe it in a sentence, drop in two examples of the output you want. Kivi shows what it took from them.

Kivi — New mode16:38

NAME

Interview mode

SAY WHAT IT SHOULD DO

"When I'm practising, take whatever I ramble and turn it into a STAR answer. Keep my real numbers. Don't make me sound like a brochure."

⏮

spoken, 14 seconds

EXAMPLES OF GOOD OUTPUT

2 added

"Our onboarding drop-off was 38%. I ran five interviews, found the OTP step was the block, shipped a resend fix in a week. Drop-off went to 21%."

"The migration had no owner. I wrote the cutover plan, split it across three people, and we moved 40 services with one hour of downtime."

+ paste or speak another example

WHAT KIVI TOOK FROM YOUR EXAMPLES

01

Four beats: situation, task, action, result.

02

Open with a number where you have one.

03

Under 90 words. First person, past tense.

04

Never invent a metric you did not say.

DRY RUN

on a recording from yesterday

"Umm so basically there was this thing where the design team and us kept, like, redoing each other's work, and I kind of stepped in and we made a shared tracker, and it got better I think."

Design and engineering were duplicating work across two backlogs. I took ownership of the overlap, set up a single shared tracker, and ran a weekly triage. Duplicate tickets dropped to almost none within a month.

too formal

shorter

keep my hedging

Nothing is saved until you turn it on.

DISCARD

SAVE AND TURN ON

23 A mode, running

No selection, no instruction. You just answer the question and the mode does the shaping.

Notion — Interview prep

INTERVIEW MODE 17:12

PRACTICE · QUESTION 4

Tell me about a time you handled conflict between two teams.

YOUR ANSWER

Design and engineering were duplicating work across two backlogs. I took ownership of the overlap, set up a single shared tracker, and ran a weekly triage. Duplicate tickets dropped to almost none within a month.

Q3 and Q5 answered the same way. Q1 still needs a number.



Interview mode



"Umm so basically there was this thing where the design team and us kept redoing each other's work..."

situation

task

action

result - no number

You didn't give a figure for the drop. Say one and Kivi will put it in the last line.



listening for question 5

MODE OFF

24 Personal dictionary

A word Kivi got wrong is fixed where it happened, and the fix becomes an entry.

Slack – #sarvam-pilot11:26

MESSAGE

Numbers for the Sarvam pilot are in the sheet – ArJun is adding the last two districts tonight.

Teach a word2 TO CONFIRM

"No – Sarvam AI, and Arjun with a small j."

Sarvam → Sarvam AIheard 6×

ArJun → Arjuna person

Kivi will use these in every app, in English and in Tamil script, and will stop offering the old spellings.

ⓘ

JUST THIS ONCE

ADD TO DICTIONARY

Your dictionary34 entries

Sarvam	Sarvam AI
RLHF	spell as initials, never expand
amma	keep in Tamil, never "mother"
Kshatriya	surname of my co-author
sem	semester, not "same"

25 Styles, from your own writing

Point Kivi at three things you already wrote. The style is the shape it finds in them.

Kivi — Styles09:20

Styles

A style is a shape for the output. It can sit under any Intention.

NEW STYLE

IN USE

Research notes

Finding · Evidence · Limitation · My observation

Weekly update

Shipped · In flight · Blocked · Ask

Message to amma

Tamil, short lines, no jargon, ends with a question

LEARNING A NEW STYLE · WEEKLY UPDATE

THE SHAPE IT FOUND

Three updates you sent last month, chosen by you.

✓ Week 31 — Slack, #eng-weekly

✓ Week 33 — Gmail, to Prof. Raman

✓ Week 34 — Notion doc

01 Four headings, always in the same order.

02 One line per item, verb first.

03 Blocked items name a person.

04 Ends with a single ask, never a summary.

Kivi reads only what you point at. Nothing else in these apps is looked at.

Try it on this week's notes?

PREVIEW

SAVE STYLE

26 Explained aloud

The output does not have to be text. Kivi speaks the explanation while your eyes stay on the page.

Preview — paper.pdf

SPEAKING 08:04

2 · BACKGROUND

Pipelines of this kind normally begin with language identification, which fixes a decoding path before any semantic work is done.

A shared encoder maps utterances from all eleven pairs into one space, so a switch point inside a sentence does not force a re-decision about the language.

The consequence is that latency no longer scales with the number of switch points, which we verify in section 4.

Related work is discussed at the end of this section.



"The encoder puts every language into one space, so a mid-sentence switch costs nothing."

0:19 / 0:44

Tamil

English

1.25x

SAY "STOP" OR PRESS ESC

27 Language is a layer, not a setting

You never pick a language before speaking. You only ever say where the output should land.

Kivi — Language

21:02

Language

Kivi detects nothing in advance. It decides at the end, from what you asked for.

LAST UTTERANCE

"Na professor-ku mail anuppanum, but formal-ah irukanum."

HEARD

Tamil + English, 4 switch points

MEANING

write · to professor · formal

OUTPUT

English, because it is an email to Prof.
Raman

INSTEAD

Tamil · Hindi · keep as spoken

WHERE OUTPUT LANDS, UNLESS YOU SAY OTHERWISE

Gmail

English

WhatsApp

Tamil

VS Code

English, comments included

Explanations

Tamil

ACROSS THE FIVE INTENTIONS

Think

notes in the language you thought in

Write

output in the reader's language

Understand

explanation in yours, source untouched

Communicate

one thought, either language, text or voice

Remember

reminded in the words you used

22 Indian languages plus English. No language picker anywhere in the product.